



The U-Shaped Challenge of Language Grounding: How Instruction Granularity Shapes Instruction Following Agent Behavior

Anonymized Authors

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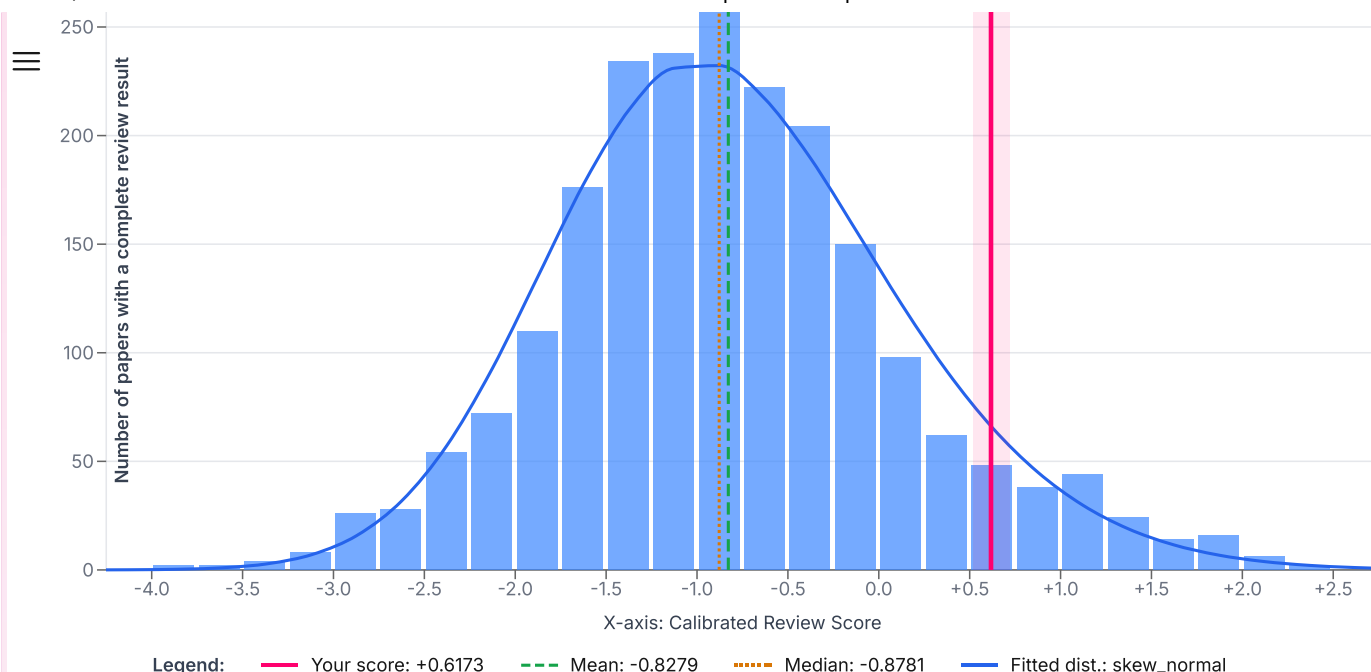
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Desk Rejection Assessment:

Paper Length

Pass ☒. The paper's length is within the typical range for an ACL main track submission.

Topic Compatibility

Pass ☒. The paper focuses on language grounding, instruction following, and embodied AI, which directly aligns with the ACL topic "Multimodality and Language Grounding to Vision, Robotics and Beyond."

Minimum Quality

Pass ☒. The paper is well-structured with all necessary sections (Abstract, Intro, Related Work, Method, Experiments, Conclusion). The scientific approach is sound, the writing is clear, and it is framed as a scholarly contribution. No significant flaws that would warrant a desk rejection are apparent.

Prompt Injection and Hidden Manipulation Detection

Pass ☒. A thorough check of the paper's content reveals no hidden prompts, invisible text, or other forms of manipulation targeting the review process.

Expected Review Outcome:

Paper Summary

This paper investigates the impact of instruction granularity on language-guided embodied agents, an area the authors argue is under-formalized. They propose a novel, agent-centric metric called rule width (\$w\$), adapted from classical planning, to quantify the decision-making burden imposed by an instruction. To study this, they introduce the Mini-Behavior-Gran benchmark, which systematically varies \$w\$ across tasks. The main findings are a non-monotonic, U-shaped relationship between granularity and agent performance, an asymmetric generalization pattern from coarse to fine-grained instructions, and a performance collapse at high width values. The authors attribute the U-shaped trend to a trade-off between the difficulty of language grounding and the predictability of instructions from visual context, which encourages visuomotor shortcuts.

Summary Of Strengths

This is a strong and insightful paper that tackles a fundamental, yet often overlooked, problem in language-guided embodied AI. Its primary strengths are:

1. **A Principled Formalism for Granularity:** The key contribution is the proposal of rule width (w) as a formal, quantifiable, and cross-task metric for instruction granularity. This is a significant step up from vague, text-centric heuristics like sentence length or entity counts. The paper compellingly argues for an agent-centric view of granularity tied to planning complexity and provides solid empirical evidence in **Figure 8** that w is a more consistent predictor of agent success than other linguistic heuristics. This formalism provides the community with a valuable tool for more rigorous analysis.
2. **Novel and Surprising Empirical Findings:** The paper uncovers a non-intuitive "U-shaped" performance curve, as shown clearly in **Figure 5**. The finding that performance is high at the fine-grained ($w=0$) and coarse-grained ($w \geq 2$) extremes, but dips in the middle ($w=1$), is a surprising result that challenges the simple assumption that coarser instructions are always harder. The discovery of a hard performance cap at $w \geq 4$ (see **Table 2**) is also a striking result, suggesting a fundamental cognitive bottleneck in current VLA models that mirrors complexity barriers in symbolic planners.
3. **Rigorous and Controlled Experimental Design:** The introduction of the Mini-BEHAVIOR-Gran benchmark is a valuable contribution in itself. By isolating instruction granularity from perceptual noise and task horizon, the authors are able to conduct a clean, controlled study. The experimental setup, including the oracle instructor and the systematic variation of width, allows for clear attribution of performance differences to the variable of interest. The analysis in **Figure 4**, which decouples width from the number of rule calls (instruction horizon), effectively demonstrates that width captures an intrinsic difficulty independent of sequential length.
4. **Insightful Mechanistic Analysis:** The paper goes beyond simply reporting the U-shaped trend. It proposes a compelling mechanistic explanation rooted in the trade-off between grounding complexity and instruction predictability ($P(l|v)$). The authors provide a chain of evidence for this: a theoretical argument for why predictability increases with coarseness, empirical validation via CE loss in **Figure 7**, and a targeted perturbation analysis in **Figure 6** that quantifies the resulting shift to "shallow grounding". This deep dive into the *why* is a hallmark of excellent scientific work.

Summary Of Weaknesses

Despite its many strengths, the paper has several weaknesses that temper its conclusions and limit the immediate generalizability of its findings.

1. **Limited Scope and Reliance on a Simplified Environment:** The most significant weakness is that all experiments are conducted within the Mini-BEHAVIOR environment. While the authors justify this choice for control and fast iteration (Appendix B), it is a grid-world with discrete actions and simplified physics. This raises major questions about whether the observed U-shaped trend, the width cap, and the asymmetric generalization are fundamental properties of language grounding or artifacts of this specific, highly-structured environment. The translation from raw observations to a symbolic feature set Φ is crucial for the width calculation, but how this process would scale to more complex, realistic environments with continuous states and noisy perception is not addressed.
2. **Over-reliance on an Oracle Instructor:** The experimental design (visualized in **Figure 2**) uses an oracle instructor to provide the agent with the correct rule r_t at each step. While this isolates the single-step grounding problem, it removes the agent's need to maintain a high-level plan or even track which instruction it is currently following. This setup may oversimplify the agent's task. In a more realistic scenario, an agent receiving a coarse instruction (e.g., "clean the shoes") would need to internally decompose it and track its progress, a challenge that is completely offloaded to the oracle here. The paper's argument in the Limitations section (Page 9) that the grounding gap remains is valid, but it sidesteps the fact that a major part of dealing with coarse instructions—task decomposition—is removed from the agent's responsibility.
3. **The "Shallow Grounding" Claim Could Be Stronger:** The diagnosis of shallow grounding is primarily based on the ΔSR_{w-w_0} metric from **Figure 6**. While the smaller performance drop for coarse-trained models is indicative, this is an indirect, behavioral measure. The claim could be substantially strengthened with more direct probes. For instance, an analysis of the model's internal representations (e.g., attention maps) could show whether coarse-trained agents learn to ignore the

language input and focus only on specific visual cues. Without this, "shallow grounding" remains a plausible but not definitively proven explanation.

4. **Unclear Origins of the "Width Cap":** The paper reports a hard performance collapse at $w \geq 4$ (Table 2) and links it to the $O(|\Phi|^k)$ complexity of symbolic planners. This is a fascinating observation, but it is presented as a fixed barrier. Is this cap truly fundamental, or is it a function of the specific model size (Prismatic-7B), architecture, or the amount of training data? It's plausible that a larger model or more diverse training could push this boundary. The paper would be stronger if it explored this limit, for instance, by testing smaller models to see if the cap appears at a lower width.
5. **Asymmetric Generalization Explanation is Incomplete:** The paper finds that coarse-trained models generalize to fine instructions but not vice-versa (Table 2). The explanation given is that coarse-training creates a "visual-dominant policy" that is robust to changes in language. This makes sense. However, the failure of fine-to-coarse generalization is explained as the policy being "brittle when detailed guidance is withdrawn." This is a bit circular. A deeper explanation is needed. For example, does the fine-grained model fail because it never learned the necessary long-horizon visual-motor skills that are implicitly taught during coarse-grained training? The coarse trajectories are inherently longer and more complex, potentially providing a richer learning signal that the short, segmented fine-grained trajectories lack.
6. **Potential Confounding of Width and Task Complexity:** While the paper attempts to decouple width from horizon, it's not entirely clear if width is fully decoupled from the inherent complexity of the sub-task encoded by the rule. For example, in Table 1, cleaning the second shoe has $w=4$, which requires tracking everything from before plus the state of the first shoe. This seems more complex than just a higher "concurrent feature" count. A rule with $w=4$ might correspond to a sub-task that is simply harder to execute for reasons beyond just feature tracking (e.g., longer action sequence, more precise navigation). The paper should discuss this potential confounder more directly.
7. **Missing Critical Related Work:** The literature review misses several highly relevant papers that deal with instruction granularity, abstraction, and grounding. This oversight weakens the positioning of the paper's contribution. (See next section for a detailed list).

Potentially Missing Related Work

The paper's literature review is solid in some areas but misses several key contemporary and foundational works on instruction granularity and language grounding, which should be incorporated to better contextualize the contribution.

- Körner, A., Castillo, M., Drijvers, L. (2023), *Embodied Processing at Six Linguistic Granularity Levels: A Consensus Paper*. This is a recent review paper that explicitly discusses embodied language processing across different levels of linguistic granularity. It provides a strong conceptual background that this paper would benefit from engaging with, particularly in the Introduction and Related Work sections.
- Cohen, V., Tasse, G. N., Gopalan, N. (2024), *Compositional Instruction Following with Language Models and Reinforcement Learning*. This recent paper tackles compositional instruction following, a problem deeply intertwined with granularity. It combines LLMs and RL to handle complex instructions and should be discussed as an alternative approach to managing instructional complexity in Section 2.
- Feng, W., Fu, T.-J., Lu, Y. (2022), *ULN: Towards Underspecified Vision-and-Language Navigation*. This work introduces a benchmark for evaluating agents on underspecified instructions of varying levels. This is directly related to the concept of granularity and should be cited and compared with the Mini-BEHAVIOR-Gran benchmark in Section 4.
- Burns, K., Manning, C. D., Fei-Fei, L. (2021), *Neural Abstractions: Abstractions that Support Construction for Grounded Language Learning*. This paper proposes a model that learns abstractions to generalize to new instructions, which relates to how agents might handle different granularities. It's relevant for the discussion on generalization in Section 5.5.
- Yu, H., Zhang, H., Xu, W. (2018), *Interactive Grounded Language Acquisition and Generalization in a 2D World*. This is an earlier but relevant work on how agents learn grounded language from scratch in a 2D world, touching upon generalization from different types of instructions. It would provide useful context in Section 2.

Comments Suggestions And Typos

The paper is well-written, but dense. Here are some suggestions for improvement and questions that could clarify key points. My score could increase if the authors can convincingly address the concerns about the

generality of their findings.

1. **Strengthening the "Shallow Grounding" Claim:** As mentioned in the weaknesses, the evidence for shallow grounding could be stronger. I would encourage the authors to include an analysis of the model's internal states. For example, do attention maps for the Pure-C model show that it largely ignores the text tokens and focuses on visual objects, compared to the Pure-F model? This would provide much more direct evidence for the claimed mechanism.
2. **Exploring the Width Cap:** To make the $w \geq 4$ capacity cap claim more robust, it would be valuable to see if this cap is sensitive to model scale. Training a smaller VLM (e.g., a 1B parameter model) on the same data could reveal if the cap appears earlier (e.g., at $w \geq 3$). This would help disentangle whether the cap is a fundamental cognitive limit or a resource-dependent one.
3. **Generalizability Discussion:** While running experiments in a more complex environment is likely out of scope for a rebuttal, the paper would be significantly improved by a more detailed discussion on the path to generalization. How would the rule width formalism apply in environments where a clean symbolic feature set Φ is not available? Would one need to train a perception module to produce these features? How would that module's errors affect the analysis? A thoughtful discussion in the Limitations or Conclusion would strengthen the paper.
4. **Clarity on Rule Formulation:** In Section 3.3, the paper states rules are of the form $C_{r_i} \mapsto E_{r_i}$. Could the authors clarify if the rules are manually defined for each of the 20 tasks, or if there is a semi-automated process? Understanding the design effort for the rule sets is important for assessing the scalability of this benchmark-creation methodology.
5. **Minor Typos/Clarifications:**
 - The citations in the reference list seem to have some formatting inconsistencies (e.g., "Bontine et al. (2021)" and "Bontine and Gartner (2021)" should likely be "Bonet"). Please double-check the bibliography.
 - In **Figure 4**, the 'nan%' cells are a bit distracting. It might be clearer to just leave them blank or shade them grey, with a caption explaining that no data exists for those bins.

My overall assessment could be raised if the authors can provide stronger evidence for their mechanistic claims (e.g., via attention analysis) and offer a more convincing argument or preliminary experiment regarding the generalizability of their findings beyond the current simplified setting.

Confidence

5 Completely confident in evaluation: Positive that my evaluation is correct. I read the paper very carefully and am familiar with its content.

Soundness

4 Strong: This study provides sufficient support for all of its claims. Some extra experiments could be nice, but not essential. (Justification: The study is very well-controlled and the claims are well-supported *within the defined experimental setup*. The soundness is strong for the specific questions it asks, though the broader applicability is a separate question.)

Excitement

4.0 Exciting: I would mention this paper to others and/or make an effort to attend its presentation in a conference. (Justification: The formalization of granularity and the discovery of the U-shaped curve are genuinely novel and thought-provoking. This work opens up a new, more rigorous way to analyze instruction-following agents.)

Overall Assessment

 [See how you rank ↑](#)

4 Conference: This paper could be accepted to an ACL conference. It demonstrates sufficient novelty, soundness, and impact. (Justification: The paper makes a clear, novel, and significant contribution by formalizing instruction granularity and uncovering a surprising empirical phenomenon. The analysis is thorough and the custom benchmark is a valuable resource. While there are limitations regarding generalizability due to the simplified environment, the work is a strong, self-contained piece of research that

advances our understanding of language grounding in embodied agents. It is well-suited for the main conference.)

Best Paper Justification

N/A

Limitations And Societal Impact

The authors have included a dedicated "Limitations" section (Section 7) that is commendably transparent. They correctly identify the use of discrete action spaces and the oracle instructor setup as key limitations. The discussion is honest and contextualizes the work appropriately, acknowledging that these choices were made to isolate the variable of interest (granularity) from confounds like continuous control noise. They also correctly point out that even with the oracle, the grounding task itself remains non-trivial, as evidenced by the performance collapse at high width.

One area for improvement would be to expand on the societal impact. While the work is foundational, any progress in instruction-following robots has downstream implications. For instance, robots that are robust to different instruction granularities could be more accessible to non-expert users. Conversely, the finding that agents can resort to "shallow grounding" and visuomotor shortcuts highlights a potential failure mode: a robot might appear to understand a command but fail catastrophically if the visual context deviates slightly from its training distribution. A brief discussion of these potential positive and negative societal impacts would make this section more complete.

Ethical concerns

None. The research is conducted in a simulated environment using publicly available models and datasets. The work does not involve human subjects, private data, or dual-use technologies that pose an immediate ethical risk. The authors have submitted the Responsible NLP Checklist and their work adheres to standard ethical guidelines for AI research.

Needs Ethics Review

No.

Reproducibility

4 Mostly reproducible: They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method. (Justification: The authors provide extensive details in the appendix, including hyperparameters, model architectures, and a description of the benchmark generation process. They also state the benchmark and code will be released, which would move this to a '5'. Without the code release at submission time, it is 'Mostly reproducible' based on the detailed paper.)

Datasets

4 Useful: I would recommend the new datasets to other researchers or developers for their ongoing work. (Justification: The Mini-BEHAVIOR-Gran benchmark is a valuable resource. It enables the first controlled studies on instruction granularity. Other researchers studying language grounding, robustness, or generalization in embodied agents would find this dataset very useful for targeted experiments.)

Software

4 Useful.: I would recommend the new software to other researchers or developers for their ongoing work. (Justification: Assuming the codebase for training the agents and using the benchmark is released as implied, it would be highly useful for anyone wanting to build upon this work or test their own agent architectures on this new benchmark.)



Disclaimer: Only the main paper is reviewed; anonymity and formatting are not checked. The desk-rejection assessment evaluates content completeness, correctness, novelty, clarity, and scientific quality in line with common ACL review guidelines.

Reviewed on March 11, 2026  Gemini-2.5-Pro

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